Interactive Services

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Introduction

- **Interactive services** in the sense that you send and asynchronously receive data from other users, consequently the user interface would have to be updated asynchronously.
- Also known as: **social networking services**, where the server only stores data, minimum processing is required at the server, hence the provisioning by RESTful web services is the appropriate approach.
- Examples:
  - Message Wall Service (Facebook)
  - Instant Messaging Service (WhatsApp or Twitter)

Introduction

- In order to asynchronously receive data, there are two alternatives:
  - **Polling**: the client periodically requests to the server if there is any new data for it.
  - **Pushing**: the server sends new data to the client when available.

Technological challenges

- For the **polling** strategy there is no challenge, the client sets up a new connection and requests for new data each time.
- For the **pushing** strategy there are two possibilities:
  - The client also takes the role of server, so it opens a TCP port waiting for new connections from the server.
  - The client connects to the server and keeps that connection alive forever, or until it is cut off.

Technological challenges

- For the pushing strategy, that means that:
  - A **ServerSocket** is instantiated and an auxiliary thread is blocked accepting new connections, which is, following the stub/skeleton pattern, the programming of a skeleton component within the client app.
  - The client uses **WebSockets** or similar (plain Sockets or a Push Service from a third provider), to permanently keep an open connection with its server.

WebSocket versus RESTful

- Both could represent opposite philosophies on programming web applications.
- **WebSockets** are based on using permanent connections with their servers.
- **RESTful** web services are based on using occasional connections with their servers, besides, the servers don't keep any state of the conversation with their clients.
- Social networking apps are based on **combining** both technologies for their implementation.